Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the primary goal of Online Analytical Processing (OLAP)? 1 / 1 point

- (selected) To provide fast answers to multi-dimensional analytical queries
- To analyze correlations between events in data logs
- To count and aggregate transactional data on the fly
- To discover regularities and relationships in large data sets

Nice work

Correct! As mentioned in the first lecture on data mining functionalities, the primary goal of OLAP is to process and provide fast responses to complex multi-dimensional queries, making it suitable for business intelligence tasks.

2. What is the primary difference between classification and clustering in data mining? 1 / 1 point

- (selected) Classification assigns data objects to predefined classes, while clustering groups data objects without predefined labels.
- Clustering assigns objects to predefined classes, while classification uses unsupervised learning.
- Both classification and clustering require labeled data to function.
- Classification generates numerical outputs, while clustering generates categorical outputs.

Nice work

Correct! As discussed in our second lecture on data mining functionalities, classification relies on predefined classes (e.g., spam or not spam), while clustering organizes objects based on their similarity without requiring predefined labels.

3. Which statement about clustering is accurate? 1 / 1 point

- Clustering is always hierarchical, grouping smaller clusters into larger ones.
- Clustering always generates a single unique group for each data object.
- Clustering relies on supervised machine learning to train the model.
- (selected) Clustering does not require predefined labels for the input data.

Nice work

Correct! Clustering is an unsupervised learning task that groups data objects based on their similarities without requiring predefined class labels.

4. Which of the following is an application of ranking in data mining? 1 / 1 point

- Grouping products into clusters based on user preferences.
- (selected) Google's PageRank algorithm used for ranking web pages based on links.
- Debugging errors in a codebase.
- Identifying anomalies in ECG data to detect diseases.

Nice work

Correct! PageRank analyzes links between web pages to determine their relative importance, as explained in the lecture.

5. Which of the following is an example of predictive functionality in data mining? 1 / 1 point

- Pattern Recognition
- Clustering
- (selected) Classification
- Association Rule Mining

Nice work

Correct! Classification is a predictive functionality used to assign labels to data points based on prior knowledge. It supports decision-making.

6. Which of the following is not a data representation type mentioned in the materials? 1 / 1 point

- Graphs/Networks.
- Vectors/Matrices.
- Sequences.
- (selected) Neural Networks.

Nice work

Correct! Neural networks are not a data representation type but rather a machine learning technique.

7. Which of the following best describes "Spatiotemporal Data"? 1 / 1 point

- (selected) Data that evolves over both space and time.
- Real-time data streams used for processing sensor data.
- Data arranged in rows and columns.
- A sequence of data points indexed by time.

Nice work

Correct! Spatiotemporal data capture how data changes across locations and time, such as weather patterns.

8. What happens if the dimensions are shuffled in a vector representation? 1 / 1 point

- The representation changes to an item set.
- It has no effect as vectors are order-independent.
- (selected) The data representation becomes invalid as dimensions must align with their values.
- The dimensions are reassigned automatically.

Nice work

Correct! The alignment between dimensions and their corresponding numerical values is crucial in vector representation.

9. What is a defining feature of matrix representation in data? 1 / 1 point

- (selected) Matrices are collections of vectors with the same dimensions.
- Matrices can only represent numerical data.
- Matrices disregard the order of attributes.
- Matrices can represent vector data with varying dimensions.

Nice work

Correct! A matrix represents multiple data objects where each vector shares the same dimensions, like user ratings for products.

10. Which Python data structure best models DNA sequences? 1 / 1 point

- (selected) Sequence data structure (e.g., list or string with positions).
- Dictionary
- Set
- Bag of words

Nice work

Correct! DNA sequences are inherently ordered, with each base (A, T, C, G) occupying a specific position. Sequence data effectively captures this structure.

11. What is the primary distinction between time series data and sequence data? 1 / 1 point

- Sequence data includes time points, but they are not ordered.
- (selected) In time series data, actual time points are associated with each measurement, whereas in sequence data, only the ordering of elements matters.
- Time series data has categorical values, whereas sequence data has numerical values.
- Sequence data requires a finer granularity than time series data.

Nice work

Correct! Time series data explicitly associates each measurement with a timestamp, making it distinct from sequence data where only the order is significant. Revisit the lecture materials discussing the mathematical formulation of time series and sequence data for further clarification.

12. How does graph data differ from item sets and sequences? 1 / 1 point

- (selected) Graph data represents relationships between items, while sequences only represent the order of items.
- Graph data is always weighted.
- Graph data always has directional links, whereas item sets do not.
- Graph data includes only nodes without links.

Nice work

Correct! Graph data captures both entities (nodes) and their relationships (links), making it more complex than sequences, which only represent order, or item sets, which only represent a collection of items.

13. What is stream data used to represent? 1 / 1 point

- A predefined set of data with a known start and end
- A static collection of related data points
- (selected) A series of complex data objects arriving continuously
- A sequence of historical data without timestamps

Nice work

Correct! Stream data represents a continuous flow of data objects arriving over time, such as news feeds or emails. This is different from static datasets, which represent a fixed collection of objects.

14. How does stream data differ from time series data? 1 / 1 point

- Stream data and time series are essentially the same
- (selected) Stream data involves continuous object arrival, while time series often uses pre-collected data
- Stream data deals only with numerical values, unlike time series
- Time series data lacks timestamps, unlike stream data

Nice work

Correct! Time series data typically involves a fixed dataset collected over time, whereas stream data is dynamic, representing continuously arriving objects.

15. Which of the following is NOT a characteristic of stream data? 1 / 1 point

- Continuous arrival of data objects
- (selected) Known start and end points for the data stream
- Presence of timestamps for each data object
- Data objects can be simple or complex

Nice work

Correct! Stream data typically does not have a known start or end, as it continuously arrives in real time.